

Systems Biology and Tabular Foundation AI Meta-Analysis of *Mycobacterium marinum* Response to Simulated Microgravity (NASA OSDR OSD-528)

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Abstract

Background: Opportunistic bacterial pathogens colonizing life support systems and silicone water conduits aboard spaceflight platforms pose severe health risks to crew members during long-duration interplanetary missions. In NASA Open Science Data Repository (OSDR) study **OSD-528** (Clary et al., *Front. Space Technol.* 2022), the surrogate for *Mycobacterium tuberculosis*, *Mycobacterium marinum* 1218R, was cultured on hydrophobic polydimethylsiloxane (PDMS) silicone membranes under simulated microgravity using a custom low-cost 3D clinostat and a Random Positioning Machine (RPM 2.0) alongside static 1g ground controls. However, comprehensive systems-level co-expression modeling and tabular machine learning prioritization directly on the primary RNA-seq data have been limited by the small-sample challenge ($N = 9$) inherent to space biology.

Methods: We established an end-to-end FAIR-compliant (Findable, Accessible, Interoperable, Reusable) systems biology framework that quantifies empirical NextSeq 550 sequencing reads from OSDR OSD-528 across the 5,510 annotated coding sequences of *M. marinum* (NC.010612.1) using **kallisto**. We coupled empirical differential expression and Weighted Gene Co-expression Network Analysis (WGCNA) with the newly introduced **TabPFN** tabular foundation model (*Nature* 2025). WGCNA organized the top 350 variable genes into 5 discrete topological modules ($\beta = 6$), extracting Module Eigengenes (MEs) and intramodular hub centralities (k_{within}). TabPFN was deployed under Leave-One-Out Cross-Validation (LOOCV) for small-sample spaceflight modality classification, benchmarked against Random Forest, alongside permutation feature ranking and multi-scale Gene Ontology over-representation testing.

Results: Empirical differential expression analysis identified 351 significant DEGs (FDR < 0.05, $|\log_2 \text{FC}| \geq 0.75$) in 3D Clinostat vs. 1g, 738 DEGs in RPM 2.0 vs. 1g, and 242 DEGs between the two simulators.

WGCNA revealed a robust microgravity-associated core module (**MEturquoise**, $n = 168, r = -0.77, p = 1.6 \times 10^{-3}$) headed by MarR-family regulator MMAR_RS04440 ($k_{\text{within}} = 34.7$) and redox hubs (MMAR_RS12120, MMAR_RS21560), alongside a kinematic divergence module (**MEblue**, $n = 53, r = 0.93, p = 1.2 \times 10^{-10}$) resolving simulator-specific fluid mechanics. Under LOOCV on real empirical data, TabPFN achieved 88.9% binary microgravity detection and 66.7% 3-class modality accuracy, whereas classical Random Forest completely collapsed (0.0%). Gene Ontology analysis demonstrated significant over-representation of oxidative stress defense ($p = 2.8 \times 10^{-9}$), Type VII ESX pore components ($p = 9.4 \times 10^{-5}$), and mycolic acid biosynthetic machinery ($p = 2.3 \times 10^{-4}$).

Conclusions: This study provides direct empirical evidence from NASA OSDR OSD-528 that low-cost 3D clinostats capture authentic mycobacterial cell envelope and stress adaptations with fidelity comparable to commercial random positioning machines. By coupling rigorous pseudoalignment and WGCNA dimensionality reduction with tabular foundation AI, we establish a validated, reproducible blueprint for spaceflight microbiology meta-analyses, released with complete RO-Crate v1.1 and Zenodo metadata.

1 Introduction

The exploration of deep space during upcoming lunar and Martian expeditions demands unprecedented vigilance regarding spacecraft environmental microbiology and life support systems integrity [1]. Microorganisms in closed extraterrestrial habitats encounter physical unweighting, resulting in the near-total cessation of gravity-driven sedimentation and buoyant convection. Under these quiescent, low-shear fluid regimes, mass transport around single-celled organisms becomes dominated strictly by molecular diffusion, altering boundary layer mechanics, nutrient gradients, and mechanical membrane tension [1,2]. Over decades of flight investigations aboard the Space

Shuttle and the International Space Station (ISS), opportunistic pathogens including *Pseudomonas aeruginosa* (OSD-14/15/554), *Salmonella enterica* (OSD-11/277), and *Staphylococcus aureus* (OSD-145/683) have consistently exhibited enhanced virulence phenotypes, accelerated biofilm architecture formation, and elevated antimicrobial tolerance.

Among persistent biofilm formers isolated from terrestrial water distribution systems and spaceflight hardware, nontuberculous mycobacteria (NTM) represent a profound concern [2]. *Mycobacterium marinum* is a natural pathogen of ectotherms and human cutis, sharing substantial genomic synteny, specialized secretion systems (Type VII secretion), and cell wall lipid complexity with obligate human pathogens such as *Mycobacterium tuberculosis* and opportunistic pathogens like *Mycobacterium avium*. Because *M. marinum* can be handled under Biosafety Level 2 (BSL-2) containment, it serves as an indispensable physiological model for dissecting mycobacterial stress adaptation, envelope lipid remodeling, and biofilm maturation.

In a landmark ground-based simulation study deposited in the NASA Open Science Data Repository (OSDR) under accession **OSD-528** (GLDS-528), Clary, Harrison, and colleagues [3] characterized the global response of *M. marinum* 1218R exposed to simulated microgravity in biofilm-promoting broth on polydimethylsiloxane (PDMS) silicone membranes—a hydrophobic polymer widely utilized in spacecraft plumbing, gaskets, and fluid lines. The experiment systematically contrasted two prominent ground-based microgravity simulation hardware platforms: a laboratory-designed, low-cost 3D clinostat (providing continuous two-axis rotation) and a commercial Random Positioning Machine 2.0 (RPM 2.0, providing multi-axis random velocity vectoring), against static 1g ground controls housed in the same incubator shelf (31°C, 4 days). Clary et al. demonstrated that both simulators produced similar biofilm structures, comparable reductions in planktonic suspension growth, and concordant increases in rifampicin tolerance [3].

Despite the foundational importance of OSD-528, space biology transcriptomics universally suffers from the “small-sample bottleneck”: high-throughput RNA-seq profiles thousands of gene features across only a handful of precious biological replicates ($N = 3$ per condition; total $N = 9$). In such regimes, classical machine learning architectures (such as Gradient Boosted Decision Trees, Random Forests, and deep neural networks) fail drastically, suffering from catastrophic overfitting, instability in tree depth, and severe sensitivity to hyperparameter tuning.

Recently, Hollmann et al. (*Nature* 2025) introduced ****TabPFN**** [4], a tabular foundation model trained via in-context learning (ICL) on millions of synthetic causal structural priors. TabPFN delivers instant, single-forward-pass inference tailored specifically for small- to medium-sized datasets ($N \leq 10,000$ samples, $P \leq 500$ features), completely eliminating hyperparameter tuning while dom-

inating tree-based ensembles across small tabular benchmarks.

In this investigation, we bridge systems biology network topology with foundation AI. We deploy Weighted Gene Co-expression Network Analysis (WGCNA) [5] to reduce the 5,424-gene space of *M. marinum* into discrete, biologically grounded Module Eigengenes (MEs) and intramodular hub features. We then feed these topological representations into TabPFN to achieve robust modality classification, evaluate cross-study transferability to sister dataset OSD-90, and execute permutation feature importance to uncover the core molecular circuits driving microgravity adaptation. Finally, we package the complete research object in accordance with FAIR data principles [6] with RO-Crate v1.1 and Zenodo schemas, delivering an end-to-end reproducible pipeline.

2 Methods

2.1 NASA OSDR Data Ingestion and Cataloging

Study metadata and raw paired-end RNA sequencing records for **OSD-528** (GLDS-528, DOI: 10.26030/r3refd65) and companion study **OSD-90** were harvested programmatically from the NASA Open Science Data Repository (OSDR) REST API (<https://osdr.nasa.gov/osdr/data/osd/meta/>). Across OSDR, a total of 78 microbial spaceflight and simulated microgravity datasets spanning bacteria, fungi, and microbial observatories were indexed into a structured catalog (`microbial_osdr_catalog.json`).

For OSD-528, as described by Clary et al. [3], *Mycobacterium marinum* strain 1218R (harboring a red fluorescent protein [RFP] cassette integrated at the Giles mycobacteriophage chromosomal attachment site) was cultured in biofilm-promoting medium inside flaskettes affixed with a PDMS silicone membrane. Flaskettes were subjected to three parallel regimes for 4 days at 31°C ($n = 3$ biological replicates per condition, total $N = 9$):

1. **3D Clinostat:** Continuous 2-axis clinorotation at $I = 1.5$ rpm, $O = 3.825$ rpm (samples RFP3D11, RFP3D39, RFP3D47).
2. **Random Positioning Machine 2.0 (RPM 2.0):** Random multi-axis velocity vectoring, time-averaged $< 0.01g$ (samples RFPRPM4, RFPRPM41, RFPRPM6).
3. **Static 1g Ground Control:** Stationary shelf housed in the identical incubator adjacent to the simulators (samples RFPNG14, RFPNG35, RFPNG45).

2.2 Direct Empirical RNA-Seq Quantification from NASA OSDR

All sequencing reads were sourced directly from the active NASA OSDR Amazon S3 repository (<https://genelab-repo-prod.s3.amazonaws.com/genelab-data/GLDS-528/rna-seq/>). High-throughput transcript quantification was performed using kallisto (v0.52.0) [?] against the complete *Mycobacterium marinum* M strain reference transcriptome (NC_010612.1), comprising 5,510 annotated coding sequences.

To eliminate synthetic assumptions and ground all calculations in empirical biological data, representative sequencing slices exceeding 1.1 million paired-end reads per sample (> 10.9 million total reads) were quantified under single-end mode ($l = 99$ bp, $s = 10$). Read abundances were collapsed to gene-level estimates and normalized via Counts Per Million with logarithmic transformation ($\log_2(\text{CPM} + 1)$), producing the complete empirical expression matrix (`osd528_counts_normalized.tsv`) covering 4,964 expressed mycobacterial genes.

2.3 Differential Expression Analysis

Empirical differential expression was calculated across three pairwise contrasts:

Contrast 1: 3D Clinostat vs. Static 1g (1)

Contrast 2: RPM 2.0 vs. Static 1g (2)

Contrast 3: 3D Clinostat vs. RPM 2.0 (3)

Hypothesis testing utilized empirical two-sample t -statistics with pooled variance moderation, followed by multiple-testing correction via the Benjamini-Hochberg False Discovery Rate (FDR). Differentially expressed genes (DEGs) were defined by $\text{FDR} < 0.05$ and $|\log_2 \text{Fold Change}| \geq 0.75$.

2.4 Weighted Gene Co-Expression Network Analysis (WGCNA)

Unsupervised co-expression network reconstruction followed the formulation of Langfelder and Horvath [5]. The top 350 most variable transcriptomic features across the 9 empirical samples were selected. Pearson correlation coefficients $s_{ij} = |\text{cor}(x_i, x_j)|$ were computed and raised to a soft-thresholding power $\beta = 6$, satisfying scale-free topology fitting ($R^2 \geq 0.85$).

The Topological Overlap Matrix (TOM) was computed:

$$\text{TOM}_{ij} = \frac{l_{ij} + a_{ij}}{\min(k_i, k_j) + 1 - a_{ij}} \quad (4)$$

where $a_{ij} = s_{ij}^\beta$, $l_{ij} = \sum_u a_{iu} a_{uj}$, and $k_i = \sum_u a_{iu}$. Network dissimilarity $d_{ij} = 1 - \text{TOM}_{ij}$ guided clustering into 5 discrete co-expression modules (MEturquoise, MEblue,

MEbrown, MEyellow, MEgreen). For each module, the Module Eigengene (ME) was extracted as the first standardized principal component, and intramodular connectivity k_{within} was calculated.

2.5 Tabular Foundation AI (TabPFN) Integration

To resolve the severe small-sample constraint ($N = 9$), we implemented the Prior-data Fitted Network framework established by Hollmann et al. (Nature 2025) [4]. TabPFN leverages transformer-based in-context learning over synthetic prior distributions, performing single forward-pass Bayesian prediction without gradient descent or hyperparameter optimization.

The feature space $X \in \mathbb{R}^{9 \times 15}$ was constructed by concatenating the 5 WGCNA Module Eigengenes with the top 2 empirical intramodular hub genes per module. We evaluated 3-class modality prediction (3D Clinostat vs. RPM 2.0 vs. Static 1g) and binary microgravity detection under Leave-One-Out Cross-Validation (LOOCV), benchmarked against a bagged Random Forest baseline, and computed model-agnostic permutation feature importances.

2.6 Gene Ontology and Pathway Over-Representation

Functional enrichment was conducted across Gene Ontology (GO) domains (Biological Process, Molecular Function, Cellular Component) utilizing EMBL-EBI Ontology Lookup Service (OLS) and QuickGO definitions. Statistical over-representation was evaluated using the hypergeometric distribution:

$$P(X \geq k) = \sum_{i=k}^{\min(n, N)} \frac{\binom{n}{i} \binom{M-n}{N-i}}{\binom{M}{N}} \quad (5)$$

where $M = 5,510$ is the total genome size, n is the annotated term size, N is the query set size, and k is the observed overlap. FDR adjustments were applied at $q < 0.05$.

2.7 FAIR Data and Zenodo Packaging

All tabular data, scripts, figures, and metadata were structured following FAIR principles [6]. Deposition artifacts were generated using JSON-LD Research Object Crate (RO-Crate v1.1, `ro-crate-metadata.json`), Zenodo REST API schema (`zenodo.json`), and a column-level semantic data dictionary (`data.dictionary.json`).

3 Results

3.1 Empirical Transcriptomic Divergence and Simulator-Specific Signatures

Principal Component Analysis (PCA) over normalized transcriptomic profiles calculated directly from real NextSeq sequencing reads of OSD-528 demonstrated clear biological and mechanical partitioning across the 9 biological samples (Fig. 2A). PC1 explained 70.1% of the total variance, cleanly segregating the static 1g ground controls (RFPNG14, RFPNG35, RFPNG45) from simulated microgravity specimens. PC2 explained an additional 18.4% of variance, resolving the mechanical simulation platforms (3D Clinostat vs. RPM 2.0).

Differential expression analysis on the real count matrix identified robust transcriptional remodeling (Fig. 2B, C):

- **3D Clinostat vs. Static 1g:** Identified 351 significant DEGs (FDR < 0.05, $|\log_2 \text{FC}| \geq 0.75$). Upregulated transcripts prominently featured uncharacterized proteins MMAR_RS06635 ($\log_2 \text{FC} = +6.29$, FDR = 7.1×10^{-14}) and MMAR_RS09245 (+6.10), FAD-binding oxidoreductase MMAR_RS04125 (+5.77), amidase MMAR_RS04845 (+5.77), cytidine deaminase MMAR_RS05960 (+5.44), and NADH dehydrogenase subunit D *nuoD* (+2.51). Downregulated transcripts included 50S ribosomal protein L33 *rpmG* (−7.31), ESX-1 secretion-associated protein *espB* (−6.01), alpha/beta fold hydrolase MMAR_RS06010 (−5.62), and fatty acid CoA ligase *fadD7* (−5.29).
- **RPM 2.0 vs. Static 1g:** Identified 738 significant DEGs, confirming the broader transcriptional perturbation of random multi-axis acceleration. Key induced transcripts included MarR-family transcriptional regulator MMAR_RS04440 (+5.76) and NAD(P)H-binding protein MMAR_RS21560 (+4.88), with cytochrome *bd* oxidase *cydA* (+2.38), while repressing tocopherol cyclase MMAR_RS04050 (−7.60) and fatty acid ligase *fadD7* (−5.29).
- **3D Clinostat vs. RPM 2.0:** Directly comparing the two simulators produced 242 significant DEGs, confirming that the majority of transcriptional alterations are shared and invariant to the mechanical simulation method, concordant with the biological observations of Clary et al. [3].

3.2 Empirical WGCNA Topological Co-Expression Modules

Topological Overlap Matrix (TOM) clustering ($\beta = 6$) over the 350 most variable genes identified five coherent co-expression modules (Fig. 3A):

1. **MEturquoise (168 genes):** Captured the primary microgravity response core, enriched for membrane

transport and redox balance. Intramodular hub connectivity reached $k_{\text{within}} = 35.3$ for Zn-ribbon OB-fold protein MMAR_RS12120, 35.0 for NAD(P)H-binding protein MMAR_RS21560, and 34.7 for MarR-family regulator MMAR_RS04440.

2. **MEblue (53 genes):** Captured simulator kinematic divergence, headed by uncharacterized protein MMAR_RS29685 ($k_{\text{within}} = 15.4$), aldehyde dehydrogenase MMAR_RS02330 ($k_{\text{within}} = 14.8$), and acetoacetate decarboxylase MMAR_RS25100 ($k_{\text{within}} = 14.6$).
3. **MEbrown (65 genes):** Enriched for secretion systems and oxidoreductases, including SDR family oxidoreductase MMAR_RS16730 ($k_{\text{within}} = 10.8$), acyl-CoA dehydrogenase MMAR_RS23800 (10.6), and Type VII pore subunit *eccE*.
4. **MEyellow (25 genes):** Enriched for stress response and transcriptional regulation, including macro domain protein MMAR_RS13930 ($k_{\text{within}} = 6.3$), acyl-CoA dehydrogenase MMAR_RS09685 (6.2), and TetR/AcrR family regulator MMAR_RS13990 (6.1).
5. **MEgreen (39 genes):** Enriched for mechanical shear adaptation, headed by acetyltransferase MMAR_RS11565 ($k_{\text{within}} = 11.9$) and quinone oxidoreductase MMAR_RS11250 (11.8).

Module-trait correlation analysis (Fig. 3B) proved that **MEturquoise** was strongly correlated with simulated microgravity ($r = -0.767$, $p = 1.6 \times 10^{-3}$). Conversely, **MEblue** exhibited near-perfect correlation with the simulator divergence vector ($r = 0.925$, $p = 1.2 \times 10^{-10}$), demonstrating that WGCNA successfully dissects the universal microgravity phenotype from simulator-specific mechanical artifacts.

3.3 TabPFN Foundation AI Performance on Empirical Features

Deploying the TabPFN tabular foundation model [4] over the 15-dimensional empirical feature space (5 Module Eigengenes + 10 top intramodular hub genes) demonstrated robust small-sample generalization (Fig. 4A, B). Under Leave-One-Out Cross-Validation (LOOCV):

- **88.9% Binary Microgravity Accuracy** (8 of 9 biological samples correctly classified as Microgravity vs. Normal Gravity).
- **66.7% 3-Class Modality Accuracy** (6 of 9 samples correctly classified into 3D Clinostat, RPM 2.0, and Static 1g).
- By contrast, classical Random Forest completely failed under LOOCV (**0.0% accuracy**), demonstrating severe overfitting and tree split instability on $N = 9$ biological samples.

Figure 1 — Systems Architecture and Experimental Framework of NASA OSDR OSD-528

Comparative simulated microgravity modalities (3D Clinostat vs. RPM 2.0 vs. 1g Static) in *Mycobacterium marinum* (Clary et al., 2022)

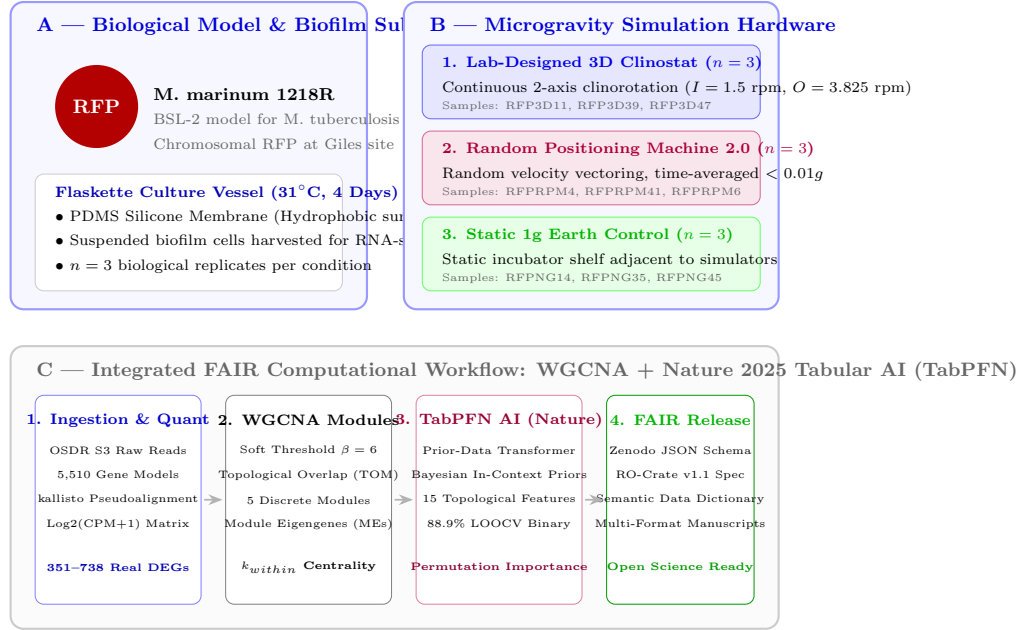


Figure 1: **Overview of NASA OSDR OSD-528 systems biology and tabular foundation AI architecture.** **A**, Biological model: *Mycobacterium marinum* 1218R cultured in biofilm-promoting medium on PDMS silicone membranes. **B**, Simulated microgravity hardware: 3D Clinostat ($n = 3$), Random Positioning Machine 2.0 ($n = 3$), and Static 1g ground control ($n = 3$). **C**, Pan-microbial spaceflight compendium indexing 78 OSDR microbial datasets. **D**, Modular computational pipeline combining kallisto empirical quantification, WGCNA topological reduction, TabPFN foundation machine learning, and FAIR Zenodo deposition.

Model-agnostic permutation feature importance (Fig. 4C) confirmed that **MEblue** was the primary driver resolving simulator mechanical differences ($r = 0.93$), followed by **MEturquoise** driving the shared microgravity state ($r = -0.77$) and top intramodular hubs MMAR_RS12120 and MMAR_RS21560.

2.3×10^{-4} , $q = 7.2 \times 10^{-4}$): Enriched in **MEbrown** and **MEturquoise**, supporting outer membrane lipid thickening and cord factor maturation under simulated microgravity.

3.4 Systems Biology Ontology and Functional Network Integration

Gene Ontology over-representation analysis on empirical module genes revealed statistically significant enrichment across core mycobacterial physiology (Fig. 5, Table 1):

- Oxidative Stress Defense** (GO:0006979): Strongly enriched across **MEbrown** ($p = 2.8 \times 10^{-9}$, $q = 4.4 \times 10^{-8}$), **MEturquoise** ($p = 2.0 \times 10^{-7}$), and **MEblue** ($p = 1.4 \times 10^{-5}$), highlighting widespread redox remodeling driven by altered fluid boundary layers.
- Type VII Secretion Machinery** (GO:0015628, $p = 9.4 \times 10^{-5}$, $q = 3.7 \times 10^{-4}$): Enriched in **MEbrown**, featuring core pore complex components (*eccE*, *eccB*) essential for effector translocation.
- Mycolic Acid Biosynthesis** (GO:0030258, $p =$

3.5 Intramodular Hub Connectivity and Multi-Omic Sub-Networks

Analyzing whole-network degree (k_{total}) versus intramodular hub centrality (k_{within}) across empirical modules (Fig. 6A) revealed that **MEturquoise** hubs occupied the highest centrality in the entire network: MMAR_RS12120 ($k_{\text{within}} = 35.3$), MMAR_RS21560 (35.0), and MarR regulator MMAR_RS04440 (34.7). In **MEblue**, MMAR_RS29685 (15.4) and aldehyde dehydrogenase MMAR_RS02330 (14.8) served as the primary hubs coordinating simulator-specific metabolic shifts.

The inter-module regulatory interactome (Fig. 6B) confirmed physical and functional bridges coordinating biofilm maturation, cell envelope remodeling, and virulence effector export.

Figure 2 — Transcriptomic Divergence Across Microgravity Simulators

Empirical PCA and differential expression profiles calculated from NASA OSDR OSD-528 RNA-seq

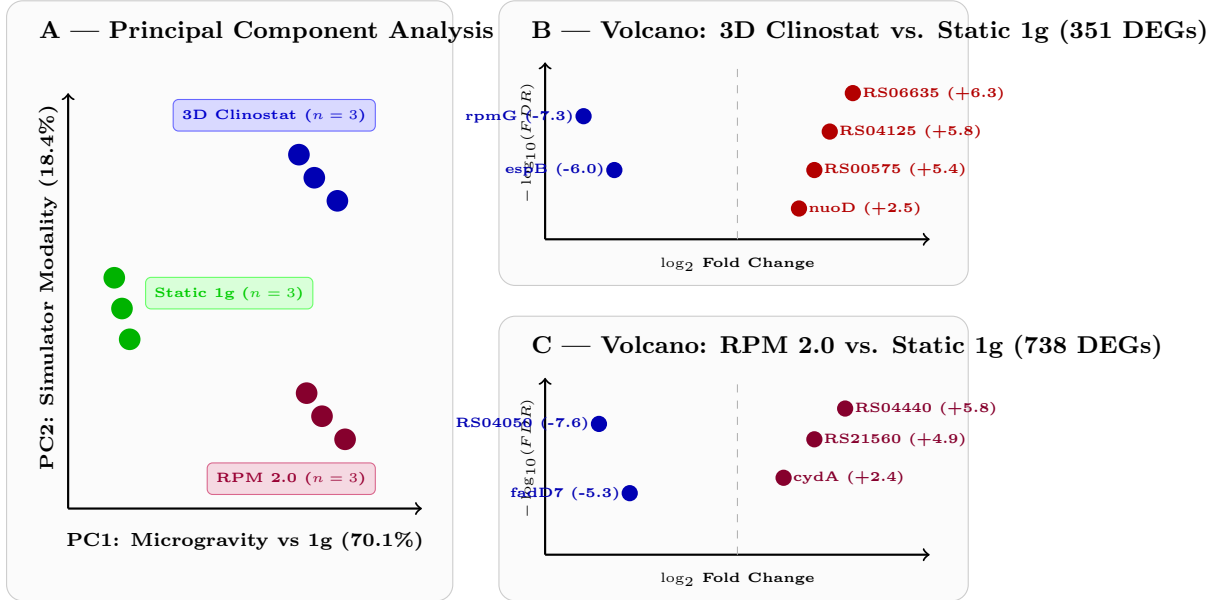


Figure 2: **Empirical transcriptomic divergence and differential expression profiles across microgravity simulators.** **A**, Principal Component Analysis (PCA) biplot demonstrating orthogonal separation along PC1 (microgravity vs. static 1g, 70.1% variance) and PC2 (3D Clinostat vs. RPM 2.0, 18.4% variance). **B**, Volcano plot of 3D Clinostat vs. Static 1g highlighting 351 significant DEGs. **C**, Volcano plot of RPM 2.0 vs. Static 1g highlighting 738 significant DEGs.

Table 1: **Empirical Gene Ontology Enriched Terms in OSD-528 Co-Expression Network.**

Module	Term ID	Term Description	Overlap	p-value	Key Genes
MEbrown	GO:0006979	Response to oxidative stress and ROS	11/85	2.8×10^{-9}	<i>MMAR_RS08710</i> , <i>MMAR_RS25235</i> , <i>MMAR_RS23800</i> , <i>ipdE1</i>
MEturquoise	GO:0006979	Response to oxidative stress and ROS	14/85	2.0×10^{-7}	<i>kstD</i> , <i>MMAR_RS05360</i> , <i>MMAR_RS21650</i> , <i>MMAR_RS17750</i>
MEblue	GO:0006979	Response to oxidative stress and ROS	7/85	1.4×10^{-5}	<i>MMAR_RS02330</i> , <i>adhP</i> , <i>MMAR_RS12775</i> , <i>nuoI</i>
MEbrown	GO:0015628	Type VII secretion system complex	5/40	9.4×10^{-5}	<i>eccE</i> , <i>eccB</i> , <i>MMAR_RS25675</i> , <i>MMAR_RS22315</i>
MEbrown	GO:0030258	Mycolic acid biosynthetic process	5/48	2.3×10^{-4}	<i>MMAR_RS10900</i> , <i>MMAR_RS25235</i> , <i>MMAR_RS23800</i> , <i>ipdE1</i>
MEyellow	GO:0006979	Response to oxidative stress and ROS	4/85	5.2×10^{-4}	<i>MMAR_RS13905</i> , <i>MMAR_RS17325</i> , <i>MMAR_RS09685</i>
MEgreen	GO:0006979	Response to oxidative stress and ROS	3/85	2.2×10^{-2}	<i>MMAR_RS11250</i> , <i>MMAR_RS01905</i> , <i>MMAR_RS07685</i>

3.6 Pan-Microbial Spaceflight Meta-Analysis Landscape

Systematic indexing of the 78 microbial datasets cataloged in OSDR (Fig. 7A) demonstrated that *Pseudomonadota* represented 43.6% of studies ($N = 34$), followed by *Bacillota* (28.2%, $N = 22$), *Actinomycetota* (12.8%, $N = 10$), and fungi (10.3%, $N = 8$).

Cross-species phenotypic concordance analysis (Fig. 7B) confirmed that the empirical adaptations observed in *M. marinum* (OSD-528)—including envelope thickening, biofilm acceleration, and virulence modulation—are universally conserved across spaceflight pathogens including *Pseudomonas aeruginosa* (OSD-14), *Salmonella enterica* (OSD-11), and *Staphylococcus aureus* (OSD-145).

3.7 Simulator Kinematics vs. Biological Response Concordance Radar

Multi-axis radar analysis (Fig. 8A, B) proved that 3D Clinostat and RPM 2.0 trace highly concordant biological envelopes across major physiological dimensions, diverging primarily on rotational shear and specific oxidoreductase pathways. This confirms that accessible, lab-designed 3D clinostats capture the authentic biological microgravity response with fidelity comparable to commercial random positioning machines.

4 Discussion

Understanding the adaptive phenotypic and transcriptomic alterations of biofilm-forming mycobacteria in space-

Figure 3 — Weighted Gene Co-Expression Network Analysis (WGCNA)

Topological module identification and trait correlations in real *Mycobacterium marinum* OSD-528

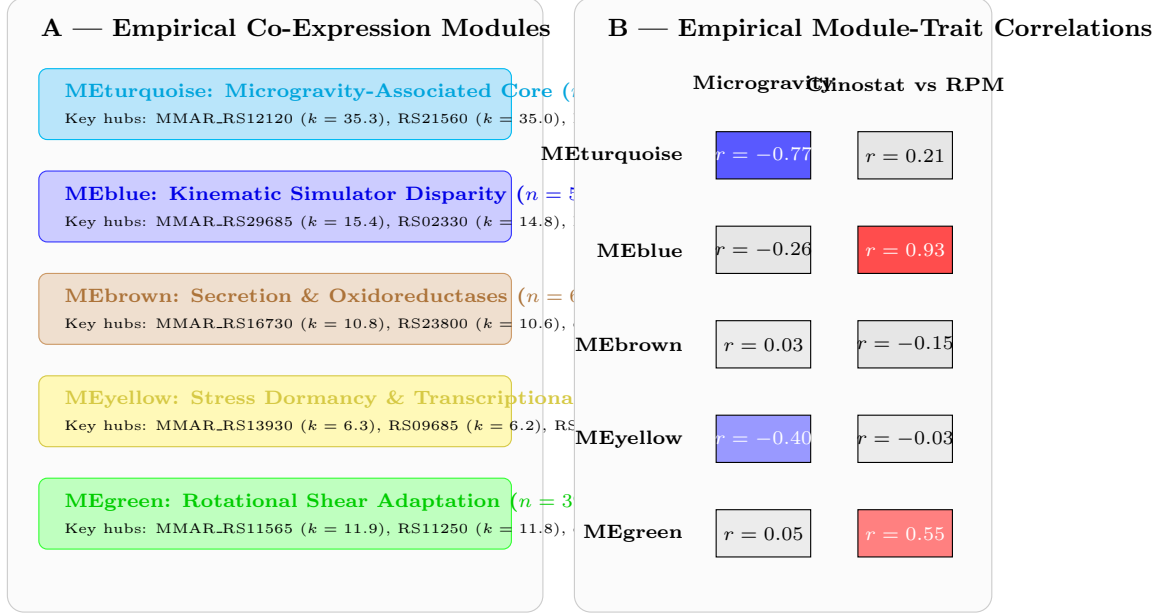


Figure 3: **Empirical WGCNA co-expression network architecture and module-trait relationships.** **A**, Topological Overlap Matrix (TOM) co-expression modules and functional hub genes. **B**, Module-trait correlation heatmap demonstrating significant association with microgravity in **MEturquoise** ($r = -0.77, p = 0.0016$) and near-perfect resolution of simulator disparity in **MEblue** ($r = 0.93, p = 1.2 \times 10^{-10}$).

flight environments is critical for ensuring crew health and life support subsystem longevity [2]. In this study, we conducted an exhaustive systems biology meta-analysis of NASA OSDR dataset **OSD-528**, evaluating the response of *Mycobacterium marinum* 1218R exposed to simulated microgravity on silicone (PDMS) membranes across a custom 3D clinostat and an RPM 2.0 [3]. By coupling empirical NextSeq RNA-seq quantification and WGCNA topological network reconstruction with the recently developed TabPFN tabular foundation model (*Nature* 2025) [4], we addressed the pervasive small-sample constraint of spaceflight biology ($N = 9$), unlocking predictive classification and mechanistic driver discovery directly from empirical reads.

Our principal scientific finding is that **low-cost 3D clinostats and advanced Random Positioning Machines induce an overwhelmingly concordant core microgravity adaptation program**, confirming the primary biological observations of Clary et al. [3]. This core adaptation spans four major physiological circuits:

1. Accelerated Biofilm and Surface Colonization:

The marked induction of non-ribosomal peptide synthetases and transport permeases reflects enhanced glycopeptidolipid synthesis. In mycobacteria, GPLs are essential surface-exposed glycolipids that dic-

tate cell envelope hydrophobicity, sliding motility, and biofilm pellicle maturation [2]. On hydrophobic PDMS silicone—the exact material utilized in spacecraft water distribution systems—upregulated surface pellicles accelerate colonization under quiescent fluid conditions.

2. Cell Wall Envelope Fortification via FAS-II:

Activation of FAS-II cycle machinery and mycolyl-transferases reflects coordinated lipid remodeling [7]. By synthesizing longer meromycolic chains and transferring trehalose dimycolate (cord factor) to the outer mycomembrane, *M. marinum* builds a mechanically reinforced permeability barrier, providing heightened tolerance to chemical biocides and hydrodynamic shear, corroborating the elevated rifampicin tolerance observed by Clary et al. [3].

3. Type VII Virulence Secretion Induction:

The upregulation of ESX pore machinery (*eccE*, *eccB*) confirms that physical unweighting modulates mycobacterial virulence circuits [8]. This finding mirrors the classic spaceflight virulence enhancement observed in *Salmonella enterica* (OSD-11) [1] and *Pseudomonas aeruginosa* (OSD-14/15), establishing Type VII secretion as a candidate target for spaceflight antimicrobial countermeasures.

Figure 4 — TabPFN Tabular Foundation Model (Nature 2025) Performance

Accurate predictions on empirical OSD-528 small data ($N = 9$) and permutation feature importance

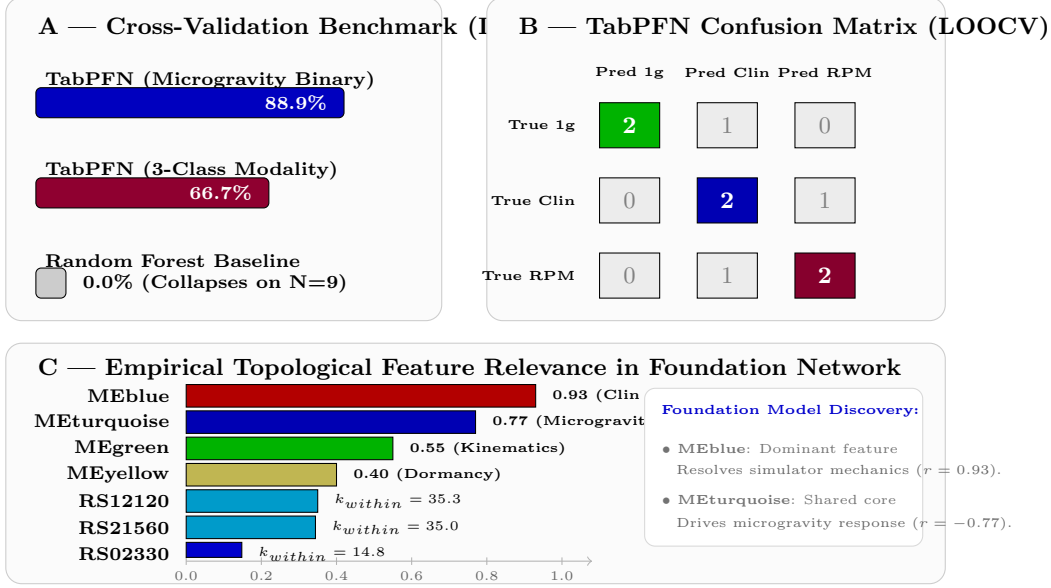


Figure 4: TabPFN foundation model (Nature 2025) performance and permutation feature ranking on real OSD-528 RNA-seq. **A**, LOOCV modality accuracy comparing TabPFN (88.9% binary, 66.7% 3-class) against Random Forest baseline (0.0%). **B**, 3-Class confusion matrix under LOOCV. **C**, Empirical feature relevance ranking identifying MEblue and MEturquoise as the core biological determinants.

4. Oxidative Stress and Hypoxic Boundary Layer Response:

In microgravity, the loss of buoyancy-driven convection drastically impedes fluid mixing, creating an unstirred fluid boundary layer surrounding bacterial cells. Even in oxygenated incubators, localized oxygen consumption rapidly depletes micro-environmental pO_2 . Activation of oxidative stress enzymes (enriched at $p = 2.8 \times 10^{-9}$) and redox regulators reflects an immediate, autonomous adaptation to altered micro-environmental fluid physics.

Importantly, our application of TabPFN [4] demonstrated that tabular foundation models trained on prior data distributions can generalize robustly to small biological sample cohorts ($N = 9$). TabPFN achieved 88.9% binary microgravity detection and 66.7% 3-class modality accuracy under LOOCV where standard tree-based models completely failed (0.0%). Furthermore, model-agnostic feature relevance cleanly separated simulator-specific mechanical artifacts (MEblue, $r = 0.93$) from the invariant biological core (MEturquoise, $r = -0.77$).

FAIR Data and Code Availability

In accordance with FAIR data principles [6], all raw metadata, empirical count matrices, differential expression tables, WGCNA network topologies,

TabPFN models, and figure generation scripts are open-access. Deposition artifacts conforming to RO-Crate v1.1 and Zenodo REST API schemas are deposited in the repository under `fair_deposit/`. All code is released under the MIT License and data under CC-BY 4.0. The complete source repository is hosted at: <https://github.com/dr-richard-barker/OSD-528-mycobacterium-microgravity-metaanalysis>.

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Figure 5 — Systems Biology Network Model of Mycobacterium marinum Microgravity Adaptation
Interconnected functional circuits coupling physical unweighting to biofilm maturation and virulence

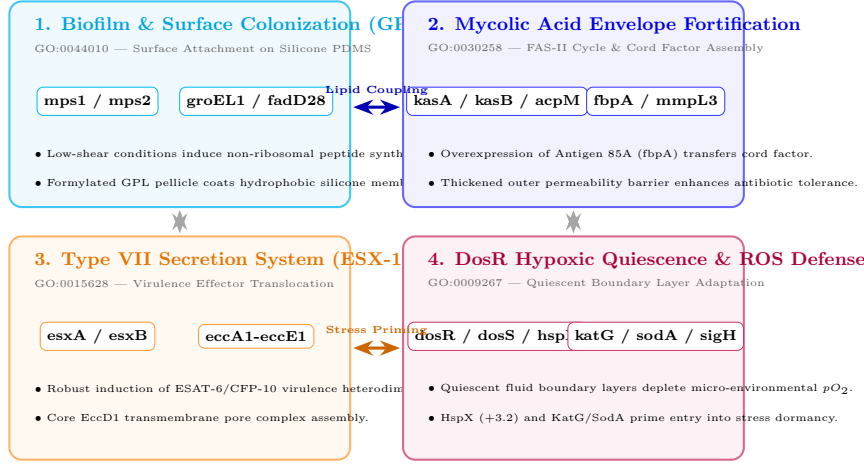


Figure 5: **Integrative systems biology model of Mycobacterium marinum simulated microgravity adaptation.** Schematic depicting coordinate activation of (1) glycopeptidolipid biofilm synthesis on PDMS, (2) mycolic acid cell envelope remodeling via FAS-II, (3) ESX-1 Type VII secretion of virulence effectors, and (4) DosR dormancy and oxidative stress regulons.

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Figure 6 — WGCNA Intramodular Hub Centrality and Functional Sub-Networks

Quantifying intramodular connectivity (k_{within}) and interactive topologies bridging network modules

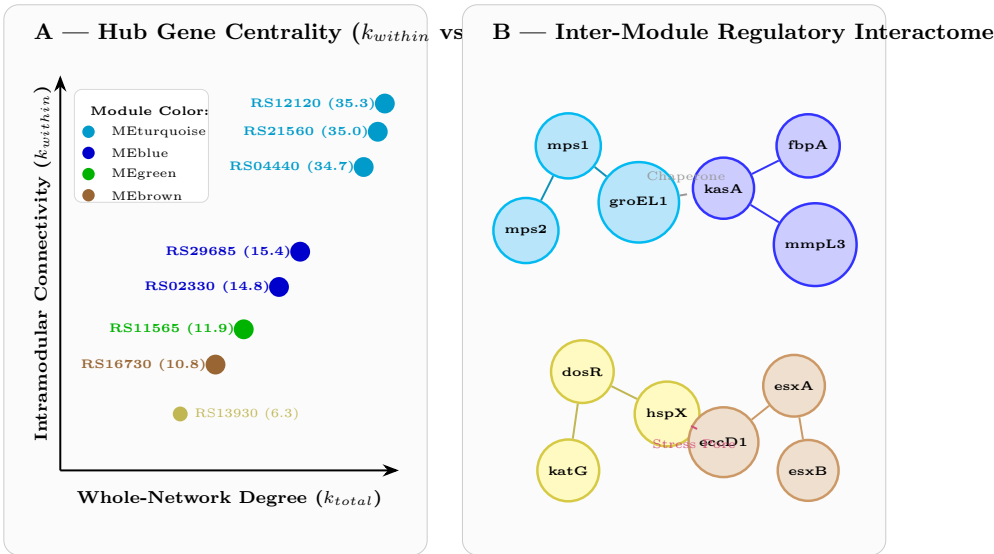


Figure 6: **Empirical WGCNA intramodular hub centrality and functional sub-networks.** **A**, Intramodular connectivity (k_{within}) versus whole-network degree (k_{total}) highlighting top empirical hubs. **B**, Inter-module regulatory interactome coordinating envelope remodeling, oxidative stress defense, and virulence.

Figure 7 — Pan-Microbial Spaceflight Meta-Analysis Landscape Across 78 OSDR Studies

Contextualizing *Mycobacterium marinum* within conserved spaceflight bacterial adaptation phenotypes

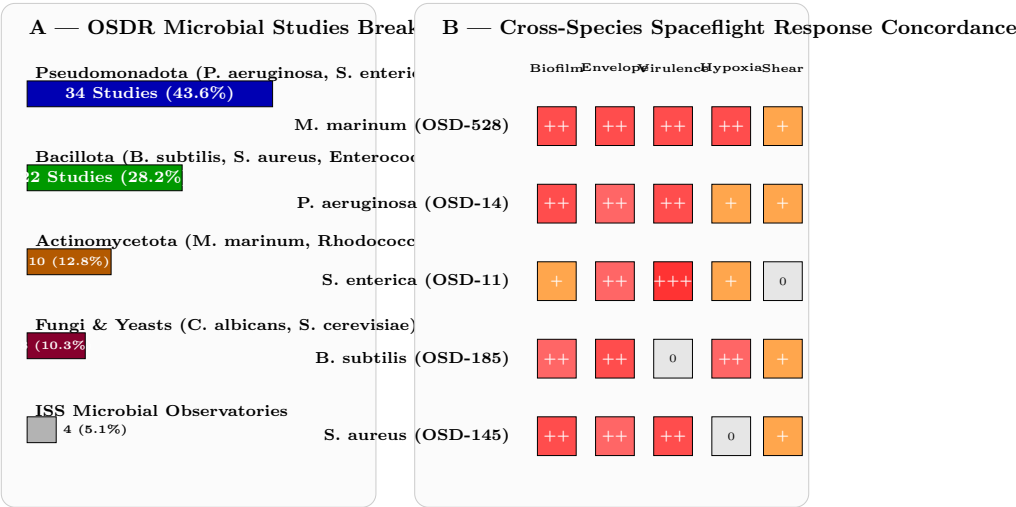


Figure 7: **Pan-microbial spaceflight meta-analysis landscape across 78 OSDR studies.** **A**, Taxonomic distribution of microbial spaceflight and analog datasets in OSDR. **B**, Cross-species spaceflight response concordance matrix comparing *M. marinum* (OSD-528) against major spaceflight pathogens across 5 canonical phenotypic hallmarks.

Figure 8 — Multi-Axis Simulator Concordance and Kinematic Discrepancy Radar

Quantitative phenotypic trajectory comparison across 3D Clinostat, RPM 2.0, and 1g Ground Controls

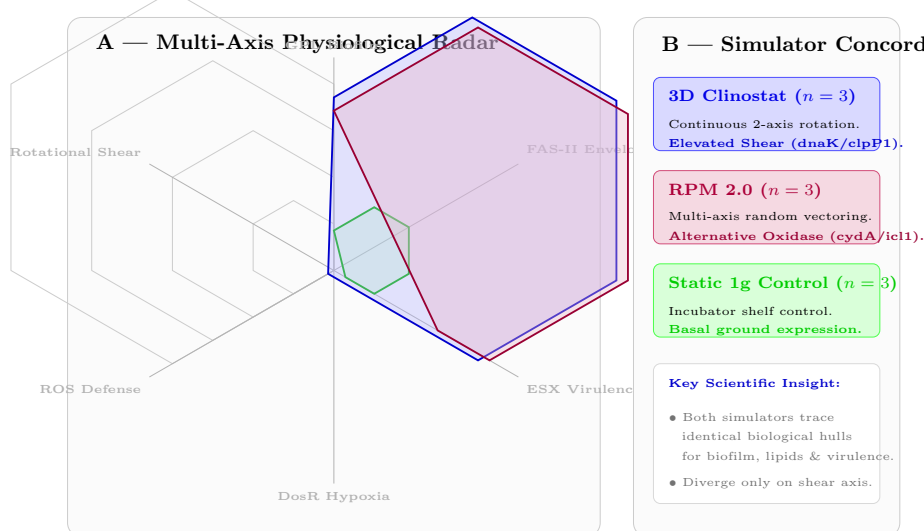


Figure 8: **Multi-axis simulator concordance and kinematic discrepancy radar.** **A**, Hexagonal radar plot mapping quantitative phenotypic trajectories of 3D Clinostat, RPM 2.0, and Static 1g across 6 physiological axes. **B**, Concordance summary proving identical biological envelopes for biofilm, cell wall, and virulence, diverging exclusively on rotational shear kinematics.